

Modeling the Survivability of Titanic Passengers Using Logistic Regression

DSC1107 - Formative Assessment 5

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I. Introduction

The Titanic was the largest ship afloat upon entering service and the second of three Olympic-class ocean liners built for White Star Line. It was a ship built to extenuate the pinnacle of luxury and comfort; equipped with many amenities on board such as a gymnasium, swimming pools, restaurants, and other accommodations. This ship the dimensions of 882 feet 9 inches (269.06 m) long, a height of 104 feet (32 m), and a maximum breadth of 92 feet 6 inches (28.19 m). The Titanic was a ship tailored for the elite, where this ship housed one of the wealthiest people at the time to experience the novelty of the ship, whilst other passengers boarded the ship were looking for a new life in the United States and Canada.

It was once stated that the ship Titanic had the most advanced technologies known at the time, this was especially true for the safety systems within the ship where it is equipped with sixteen lifeboat davits, each capable of lowering three lifeboats, for a total capacity of 48 boats. However, a tragedy occurred in the British ocean liner that sank in the early hours of 15 April 1912. Where the ship sank due to a collision with a massive iceberg and despite the equipped 48 lifeboats, a lot of passengers were sadly unable to escape the tragedy of the Titanic. As Titanic is our case study, we will be introducing a predictive model to identify passengers who have a high chance of surviving the incident, where a model called logistic regression shall be used while the data shall be obtained through the publicly available dataset from the seaborn library. Lastly, Python shall be used as the modeling procedure and shall be streamlined through libraries like Scikit Learn and Pandas.

II. Methodology

The dataset that will be used for the project is provided by the Seaborn library of Python. Once the data is loaded, standard pre-processing of the data shall be performed where initial inspections such as finding valid values and identifying leaky and unnecessary will be investigated through Pandas. This initial investigation will provide an idea of the cleaning processes where all null values are to be removed from the data set while leaky and unnecessary variables are deleted.

Since the data shall be modeled using logistic regression and another model using a hypertonic process called grid search as the target variable for the data is binary and since all values from all variables should be numerical as the model cannot interpret categorical values, this is true for both all features and the target variable, where the target variable should be only either 1 or 0 since we are using a binary version of the model and not a multi-class prediction model. For categorical variables that have a binary response such as sex, relabeling of the data shall be performed where a label of 1 shall represent the Male sex while a label of 0 shall represent the sex of the Female. Lastly, for the categorical variables that can take more than two possible values a method called one-hot encoding was performed. One-hot encoding is a process where we let the unique values of the categorical variable as their separate variables or columns, and then the values within these columns a binary data. This data represents a boolean value,

for instance, if this observation is identified to have this categorical value, it shall take a value on that variable of one, while the rest shall have a value of zero.

Another method that will be employed for the pre-processing is dealing with outliers through transformations since outliers can have a strong influence on the performance of the model, this is done through the transformation Yeo-johnson since the data contains zero values which cannot be processed by traditional methods. Furthermore, standardization of all the numerical data shall be employed since the model that we will be using the logistic regression is sensitive to scaling, to avoid this issue standardization shall be performed.

It has been identified that more observations have a value of false than true, which indicates a class imbalance within the dataset, therefore to ensure that the model is unbiased a pipeline was established where the cluster of false variables will be resampled to undersample the group, while the observations that were labeled to be true shall be resampled through the process called SMOTE which creates synthetic samples based on similar enough observations within the true group.

Lastly, the data shall be split into two datasets, with about 80% of the total dataset used for training the model and the last 20% used for testing. This is to accurately assess the model's performance and determine whether it is underfitting or overfitting, which provides insights into more advanced engineering concepts for the model, such as hyper-tuning and feature engineering.

In terms of model evaluation, the model shall be evaluated in both predicting with the train and test data through accuracy which measures the rate of the model's correct predictions over the total number of predictions. Other measurements such as precision, recall, F1-score, and confusion matrixes shall be used to create a more general performance of the model, where insight can help identify false positives and false negatives that the model makes. Lastly, AUC scores shall also be calculated to assess the model's ability to distinguish between the positive and negative classes, offering a more robust evaluation metric, particularly in cases of class imbalance. These metrics shall be applied to both the training of the model and the test period of the model to identify overfitting and underfitting behaviors.

III. Results and Discussions

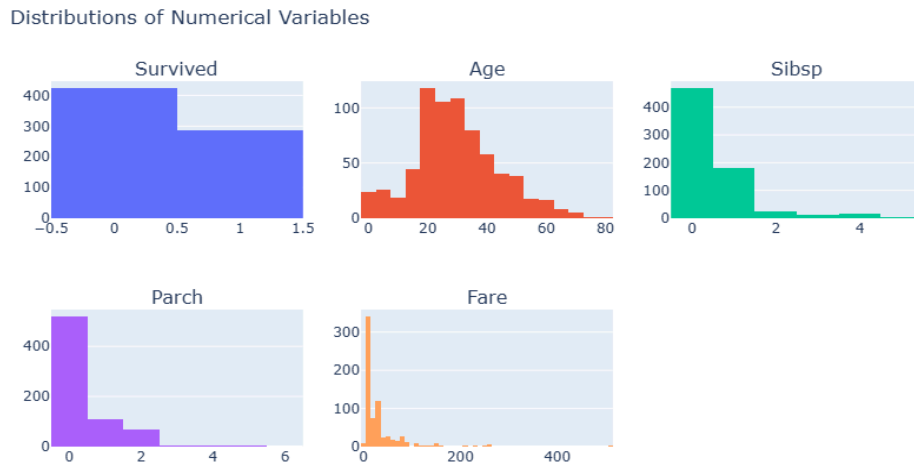


Figure 1: Histogram of all Numerical Data within the Titanic Dataset

Figure 1 plot 1 shows that there is a class imbalance within observations, where there are more observations of those that did not survive represented by the value of zero (0) in comparison to the observations that did survive represented by the value of one (1). This implies the severity that occurred within the Titanic incident, where most of the passengers were not able to survive the event.

The second plot is a plot of age, where it shows the distribution of different ages of all the passengers on board the Titanic and as observed within the plot there is a clear level of skewness. Where it is observed that more passengers are younger than those in the older age groups. This is representative of the promise of the Titanic which was to find new life for the people on board in America and Canada.

Sibsp and Parch refer to dependents on board the ship; Sibsp refers to the number of siblings or spouses an individual has along with them in the ship, while Parch refers to the number of parents on board the ship. As observed within the two plots it clearly illustrates a skew to the right where most of the passengers are either not traveling with any siblings or spouses or only with a few.

Lastly, Figure 1 of the last plot shows the fair prices that the passengers had to pay to get on board the ship. As observed this variable also takes a more skewed distribution where most of the passengers paid to be within the ship at a low fare, while there are outliers that had to pay more than that to get on board the ship.

Distributions of Object Variables

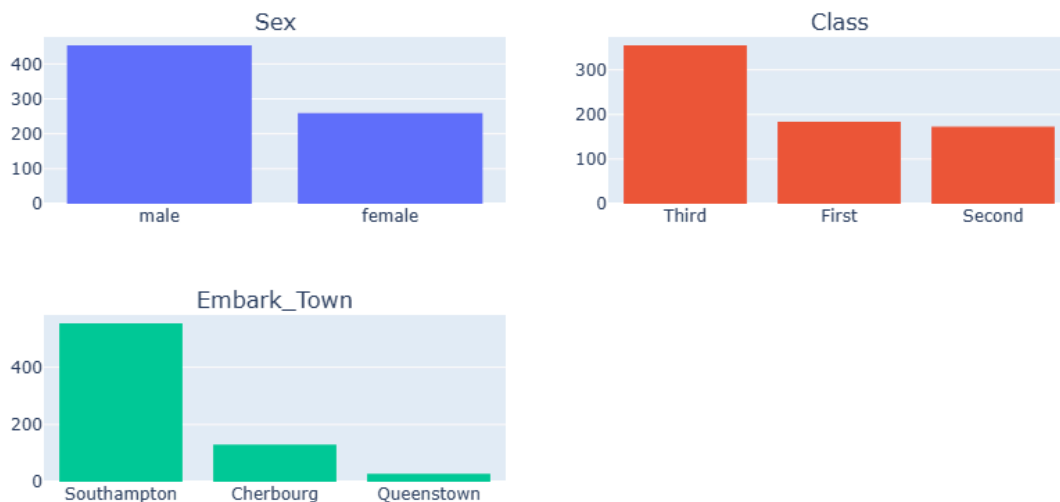


Figure 2: Counts for each Categorical Value

Figure 2 shows the counts or frequencies of each categorical value within variables that are identified to have a data type of object. As observed plot one shows that there are more male passengers than female passengers, and there are more third-class passengers than there are first or second, which implies that there are lower to middle economic-class citizens that were on board the ship rather than mid to high economic classes. Lastly, more passengers were from Southampton rather than other embarked towns.

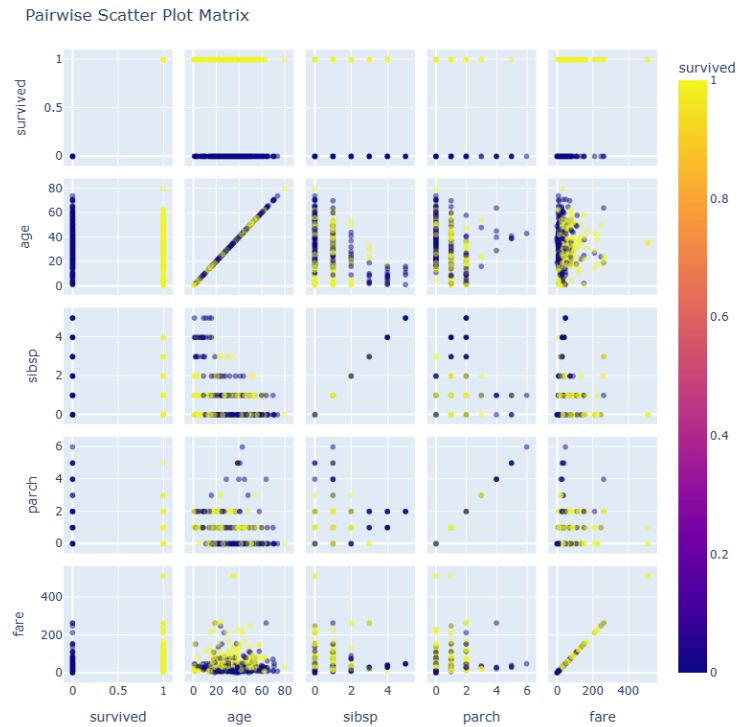


Figure 3: Pairwise plots of each numerical variable

Figure plot shows what are the relationships within each numerical variable, to which as seen in Figure 3 there is a random relationship between variables and to which it is difficult to determine the classifications of each observation. This indicates that once we calculate and identify multicollinearity using a heatmap, we will observe that there would be no strong correlation between the variables.

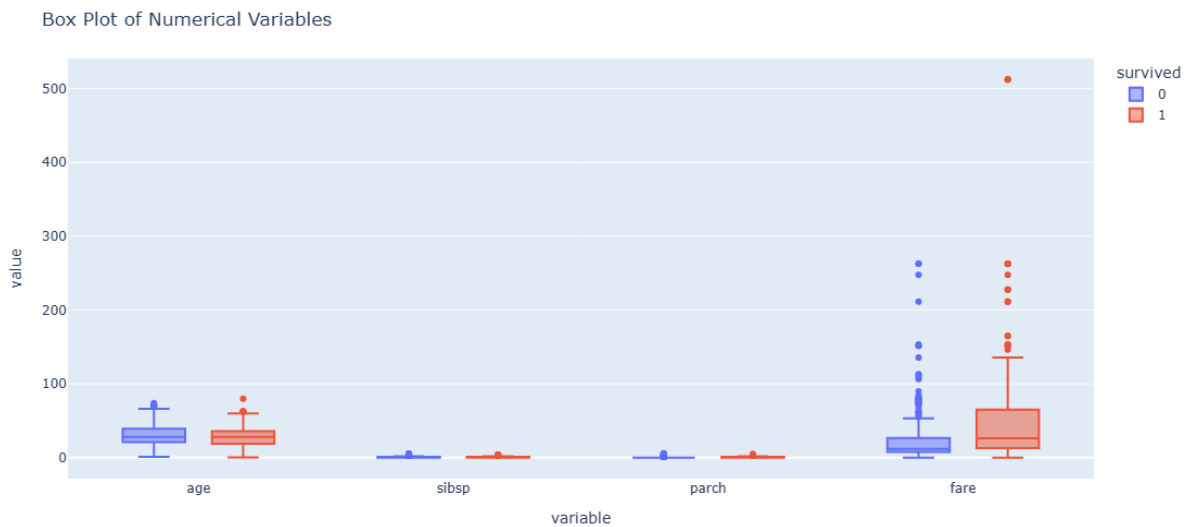


Figure 4: Boxplot of all numerical variables, grouped by survived status

Figure 4 is to represent further the distributions of the data, as can we observe that the distribution between ages for both that were able to survive and not is almost identical, while for Sibsp and Parch variables it is difficult to discern any kind of information as only low values are present for each. Lastly, the fare variables have two different distributions for both those who were able to survive and not. There is more variation of passengers that paid for different class tickets and were able to survive. On the side of those who were not able to survive, there are fewer variations with most of the data being those who paid less for their fare.

The data clearly shows that there are outliers within the data and there are tells us very different distributions which makes the model underperform. To combat this and increase the predictive power of all the variables, a method of transformation shall be used to lessen the influence of outliers. This transformation shall be done using the Yeo-Johnson transformation as the data shows that there are zero values within the dataset. Standardization shall also be used to address the scaling of the fare and ensure that all variables are comparable with each other.

Correlation Matrix Heatmap

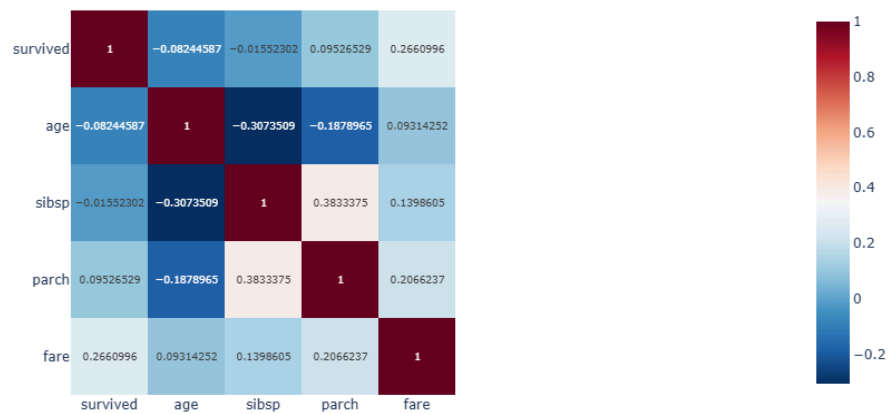


Figure 5: Correlation Matrix of all Numerical Data

Figure 5 shows that there is a correlation matrix of every pairwise variable to identify if multicollinearity is not satisfied. As observed all of the possible pairwise variables take up no correlation with each other and thus multicollinearity is not satisfied, indicating that all variables were measured independently from one another. Thus, ensuring that our model can perform well and that all variables can contribute uniquely to the prediction without redundancy, also shows that the coefficients produced by the model are highly interpretable and stable.

	model_name	accuracy	precision	recall	f1
0	Logistic Regression	0.795588	0.791304	0.802941	0.797080
1	Hypertuned Logistic Regression	0.798529	0.792507	0.808824	0.800582

Table 1: Model Evaluations on Train Dataset

Table 1 shows the evaluations of both models which are the raw logistic regression model and the hyper-tuned logistic regression using grid search. As observed the accuracy of both models is fairly similar with the logistic regression model having an accuracy of 79.56%, while the hyper-tuned logistic regression has an accuracy score of 79.85%. This implies that the hyper-tuned logistic regression model is a stronger model than the raw logistic regression, where their only difference is 0.29% which is still a big gap in the context of big data and large observations. However, we have to understand how the model balances precision and recall to say that the model produces true positives out of all actual positives. As we can observe the Hypertuned model is better at balancing precision and recall where it produces more true positives than the logistic regression with a f1 score difference of close to 1%.

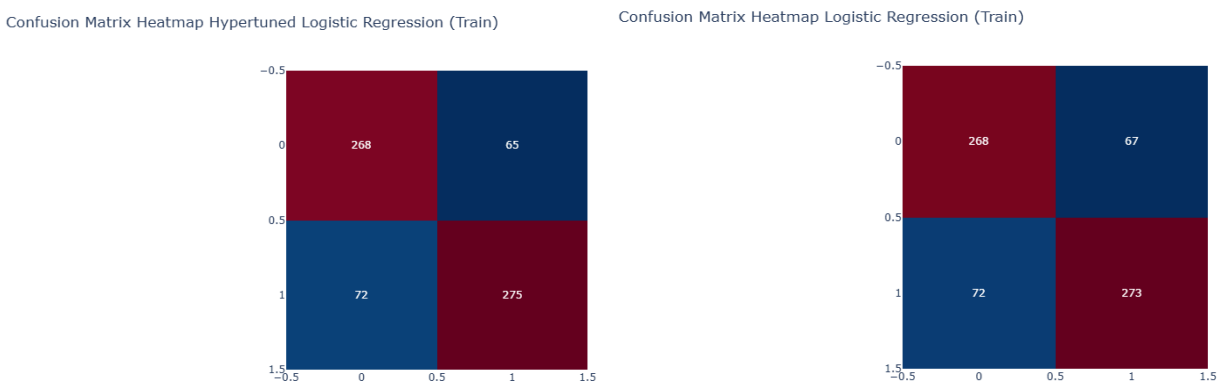


Figure 6: Confusion Matrix for Both Hypertuned (Left) and Not (Right) Logistic Regressions for Train Data

Figure 6 shows a heatmap of both the confusion matrices of the hyper-tuned logistic regression and the raw logistic regression. As shown the model performs well in the sense that there are few false negatives and false positives predicted by the model indicated by the blue color and most of them are true negatives and true positives as indicated by the red color. This implies that the model performs very well with the training data, and since the model has a few false positives and false negatives, this implies that the model is not overfitted to the data.

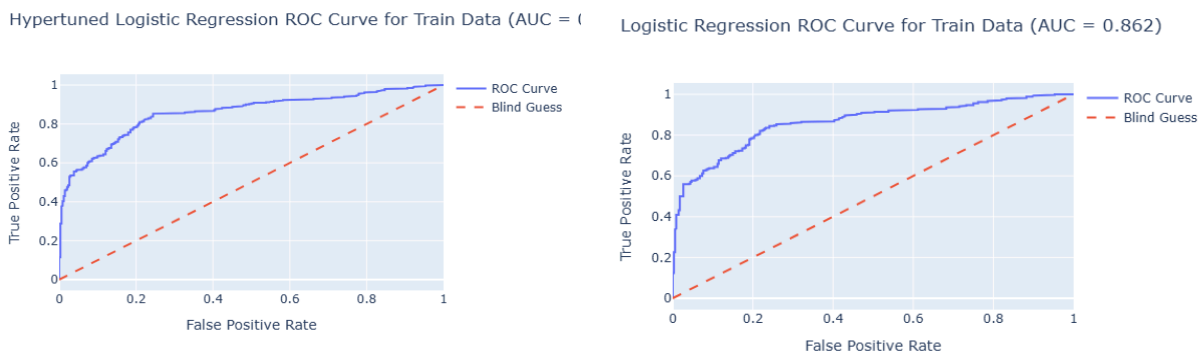


Figure 7: ROC Curve for Both Hypertuned (Left) and Not (Right) Logistic Regressions for Train Data

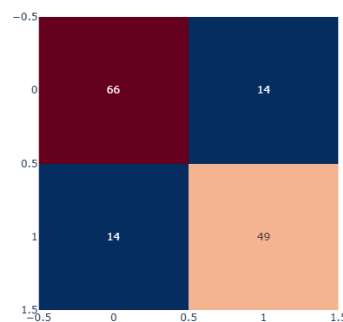
The ROC curve represents the behavior of the model when it creates a prediction of truth and communicates how many of these are true. This is represented by Figure 7 which shows that rate that shows False Positive Rate is a function of the True Positive Rate. Both models show that it takes the behavior of a log function whereas a false positive rate increases the true positive rate also increases although slowly. It is also shown that the model performs better than just blind guessing or predicting an outcome probability of just 50%. This is also communicated numerically where the hyper-tuned model has an AUC score of 0.858 while the logistic regression model has an AUC score of 0.862 which implies that both models are quite effective at distinguishing between the two classes, with the logistic regression model performing slightly better overall, and further verifies the validity of the model.

	model_name	accuracy	precision	recall	f1
0	Logistic Regression	0.811189	0.800000	0.761905	0.780488
1	Hypertuned Logistic Regression	0.804196	0.777778	0.777778	0.777778

Table 2: Model Evaluations on Test Dataset

Table 2 shows the evaluations of both models which are the raw logistic regression model and the hyper-tuned logistic regression using grid search performances within the test data. As observed the accuracy of both models has a difference of about close to 1% with the logistic regression having an accuracy score of 81.12% while the hypertuned model has an accuracy score of 80.42% this indicates that the logistic regression predicts more correct classifications than the hypertuned model. In observing the f1 score of both models we can understand how the model balances precision and recall to say that the model produces true positives out of all actual positives. As we can observe the logistic regression model is better at balancing precision and recall where it produces more true positives than the logistic regression with a f1 score difference of close to 1%. This indicates that in a real-world setting, the logistic regression is much more reliable than the hyper-tuned model.

Confusion Matrix Heatmap Hypertuned Logistic Regression (Test)



Confusion Matrix Heatmap Logistic Regression (Test)

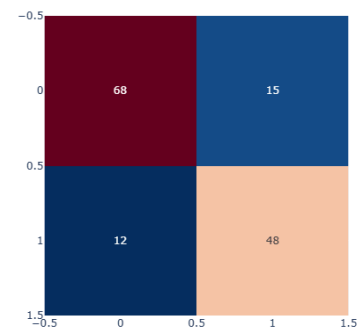


Figure 8: Confusion Matrix for Both Hypertuned (Left) and Not (Right) Logistic Regressions for Test Data

Figure 8 shows a heatmap of both the confusion matrices of the hyper-tuned logistic regression and the raw logistic regression. As shown the model performs well in the sense that there are few false negatives and false positives predicted by the model indicated by the blue color. In addition, the models predict correctly the values for true positives, however, both models find it difficult to identify true

negatives represented by the light brown color. This implies that the model performs very well with the test data, however, it finds itself not being able to predict correctly true negatives, giving more way for the model to flag false negatives.

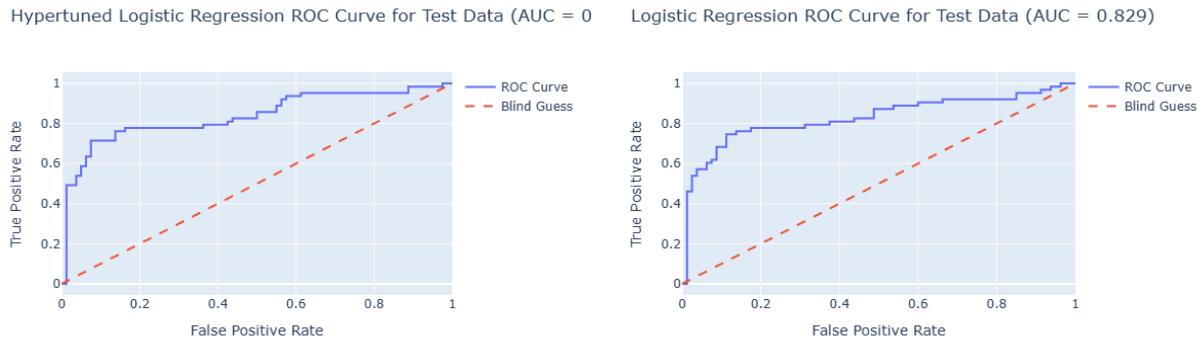


Figure 9: ROC Curve for Both Hypertuned (Left) and Not (Right) Logistic Regressions for Test Data

Figure 9 shows the ROC curve of both the hyper-tuned model and the logistic regression model where we can observe that on some intervals for the hyper-tuned model, the true positive rate is not increasing as fast in comparison to the logistic regression model. However, to truly compare both models with regards to the true positive rate calculation of the AUC score was performed with the hyper-tuned model having an AUC score of 0.838 and the logistic regression model having an AUC score of 0.829, which has about 0.01 difference between the two. However, taking into account the previously calculated F1 scores and accuracy the better model would be the logistic regression.

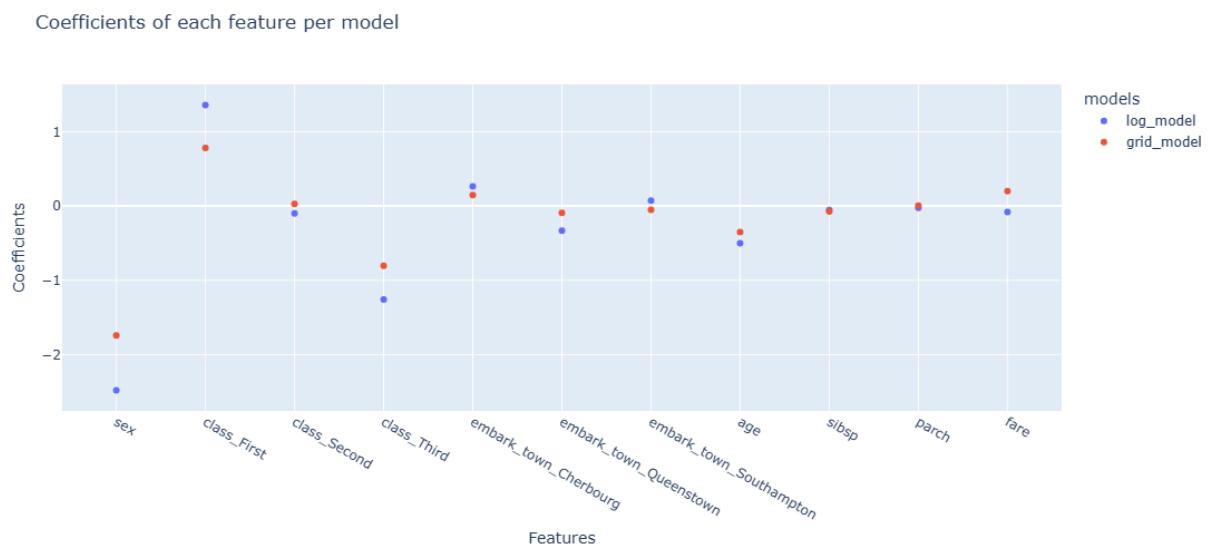


Figure 10: Coefficients for the hyper-tuned model (red) and the logistic regression model (blue)

As we can observe within Figure 10 are the coefficients that the two models calculated in predicting the classification of each observation. As shown there is little difference between the

coefficients on the features: class_second, class_third, embarked_town_cherbourg, embarked_town_queenstown, embarked_town_southampton, age, sibsp, parth, and fare. The only clear and considerable difference between the coefficients of both models are sex and the class_first variable where on sex the logistic regression model calculated a lower coefficient than the grid model being greater. In terms of the class_first variable, the logistic regression model calculated a higher coefficient than the grid model. Plot 10 shows how these models consider the importance of each feature, the more that the coefficient is near zero then the less it has for predictive power.

IV. Conclusions and Recommendations

The models were able to have an accuracy of more than 70% and have a f1 score of over 70% which indicates a moderate reliability of the models, such that it can predict the outcome of the titanic survivability. To which, as analyzed and evaluated the best model produced is the logistic regression model in comparison to the hyper-tuned model. It has a AUC score of over 0.8 indicating that the rate that the model produces correct classifications is rather high.

However this still is not enough to be considered as a high performing model as we cannot break pass the 85% percent accuracy and f1 score. Thus further pre processing procedures or more data collection should be allocated. Within the preprocessing stage the under sampling for the false labels was capped at a sampling strategy of 0.66 implying that it is the maximum possible undersampling for the group. It is recommended to explore the model's performance on raw data where there was no resampling, this is to ensure that the model is fed real world observations. Another recommendation is to explore the performance of the model if we increase the ability of SMOTE to increase the number of observations for the observations that are labeled to be true, to assess how much synthetic data can we use such that our accuracy and evaluations are larger.